

Recitation 3

ECON UN3412: Econometrics

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1 Sampling Distribution & Variance of the OLS Estimator

In Recitation 2, we showed that the Ordinary Least Squares (OLS) slope estimator $\hat{\beta}_1$ is **unbiased** ($\mathbb{E}[\hat{\beta}_1] = \beta_1$) and **consistent** ($\hat{\beta}_1 \xrightarrow{p} \beta_1$).

However, unbiasedness only tells us where the sampling distribution is centered across repeated hypothetical samples. To perform **statistical inference**—such as constructing confidence intervals and testing hypotheses like $H_0 : \beta_1 = 0$ —we must know the **spread** (variance) and the **asymptotic shape** of this sampling distribution.

1.1 The Regression Model & Least Squares Assumptions

Consider the bivariate linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n$$

where Y_i is the dependent variable, X_i is the regressor, and u_i is the error term collecting all other unobserved factors influencing Y_i .

Our analysis relies on the three core **Least Squares Assumptions (LSAs)**:

Least Squares Assumptions (LSA 1–3):

1. **LSA 1 (Conditional Mean Zero):**

$$\mathbb{E}[u_i | X_i] = 0 \implies \text{Corr}(X_i, u_i) = 0$$

The error term has a mean of zero conditional on X_i . This guarantees exogeneity (no omitted variable bias).

2. **LSA 2 (i.i.d. Sampling):**

$(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ are independently and identically distributed (i.i.d.)

This holds when data are collected via simple random sampling from the population of interest.

3. LSA 3 (Large Outliers are Rare):

$$\mathbb{E}[X_i^4] < \infty \quad \text{and} \quad \mathbb{E}[u_i^4] < \infty$$

Nonzero, finite fourth moments guarantee that extreme outliers do not dominate the sample moments, allowing us to invoke the Central Limit Theorem.

1.2 Recap: The OLS Estimation Error

Recall from Recitation 2 that the OLS formula for the slope is:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Substituting $Y_i - \bar{Y} = \beta_1(X_i - \bar{X}) + (u_i - \bar{u})$ yields:

$$\hat{\beta}_1 = \beta_1 \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sum_{i=1}^n (X_i - \bar{X})^2} + \frac{\sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Because $\sum_{i=1}^n (X_i - \bar{X})\bar{u} = \bar{u} \sum_{i=1}^n (X_i - \bar{X}) = 0$, the estimation error simplifies to:

$$\hat{\beta}_1 - \beta_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})u_i}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

1.3 Step-by-Step Derivation of $\text{Var}(\hat{\beta}_1)$

To derive the variance of $\hat{\beta}_1$ in large samples, we follow the step-by-step method presented in class:

Step 1: Normalize Numerator and Denominator by Sample Size

Divide both the numerator and the denominator of the estimation error by the sample size n :

$$\hat{\beta}_1 - \beta_1 = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})u_i}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

Step 2: Define the Composite Error Term v_i

Define the random variable:

$$v_i \equiv (X_i - \bar{X})u_i$$

(In large samples, replacing sample averages with population means gives $v_i \approx (X_i - \mu_X)u_i$.)

The numerator is simply the sample average of these v_i terms, which we denote by $\bar{v}$:

$$\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i$$

Step 3: Large-Sample Approximation of the Denominator

Consider the denominator, which is the sample variance of X (denoted s_X^2):

$$s_X^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

By the **Law of Large Numbers (LLN)** and LSA 2 & 3, as n grows large:

$$s_X^2 \xrightarrow{p} \text{Var}(X_i) = \sigma_X^2$$

In large samples, s_X^2 has negligible sampling variability and converges to the **constant** population variance σ_X^2 .

Replacing the random denominator with this constant gives the large-sample approximation:

$$\hat{\beta}_1 - \beta_1 \approx \frac{\frac{1}{n} \sum_{i=1}^n v_i}{\sigma_X^2} = \frac{\bar{v}}{\sigma_X^2}$$

Step 4: Distribution and Variance of the Numerator $\bar{v}$

Now examine the properties of $\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i$:

1. **Mean of v_i :** By LSA 1 ($\mathbb{E}[u_i | X_i] = 0$) and the Law of Iterated Expectations:

$$\mathbb{E}[v_i] = \mathbb{E}[(X_i - \mu_X)u_i] = \mathbb{E}\left[(X_i - \mu_X) \underbrace{\mathbb{E}[u_i | X_i]}_{=0}\right] = 0$$

Therefore, $\mathbb{E}[\bar{v}] = 0$.

2. **Variance of v_i :** Denote the population variance of v_i by σ_v^2 :

$$\sigma_v^2 \equiv \text{Var}(v_i) = \text{Var}((X_i - \mu_X)u_i)$$

3. **Variance of the Sample Average $\bar{v}$:** Because (X_i, Y_i) are i.i.d. (LSA 2), the terms v_i are also i.i.d. The variance of a sample mean of i.i.d. random variables is the individual variance divided by n :

$$\sigma_{\bar{v}}^2 \equiv \text{Var}(\bar{v}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n v_i\right) = \frac{\sigma_v^2}{n}$$

4. **Central Limit Theorem (CLT):** By the Central Limit Theorem, the normalized average

$\bar{v}$ is asymptotically normally distributed:

$$\bar{v} \stackrel{a}{\sim} \mathcal{N}(0, \sigma_v^2) = \mathcal{N}\left(0, \frac{\sigma_v^2}{n}\right)$$

Step 5: Applying the Variance Scaling Property

We now apply the scaling property of variances ($\text{Var}(cZ) = c^2 \text{Var}(Z)$ for any constant c).

Since σ_X^2 in the denominator acts as a fixed constant in large samples:

$$\begin{aligned} \text{Var}(\hat{\beta}_1) &= \text{Var}(\hat{\beta}_1 - \beta_1) \\ &\approx \text{Var}\left(\frac{\bar{v}}{\sigma_X^2}\right) \\ &= \frac{1}{(\sigma_X^2)^2} \text{Var}(\bar{v}) \\ &= \frac{1}{(\sigma_X^2)^2} \sigma_v^2 \\ &= \frac{1}{(\sigma_X^2)^2} \left(\frac{\sigma_v^2}{n}\right) \end{aligned}$$

Rearranging terms yields the fundamental formula for the variance of the OLS estimator:

$$\boxed{\text{Var}(\hat{\beta}_1) = \frac{1}{n} \frac{\sigma_v^2}{(\sigma_X^2)^2} = \frac{\sigma_v^2}{n\sigma_X^4}}$$

where:

$$\sigma_v^2 = \text{Var}((X_i - \mu_X)u_i) \quad \text{and} \quad \sigma_X^2 = \text{Var}(X_i)$$

Combining this with asymptotic normality gives the **asymptotic sampling distribution of $\hat{\beta}_1$** :

$$\boxed{\hat{\beta}_1 \stackrel{a}{\sim} \mathcal{N}\left(\beta_1, \frac{1}{n} \frac{\sigma_v^2}{(\sigma_X^2)^2}\right)}$$

1.4 Homoskedasticity vs. Heteroskedasticity

1. The Homoskedastic Case (Special Case)

Errors are **homoskedastic** if the conditional variance of u_i given X_i is constant across all values of X_i :

$$\text{Var}(u_i | X_i) = \sigma_u^2 \quad \text{for all } i$$

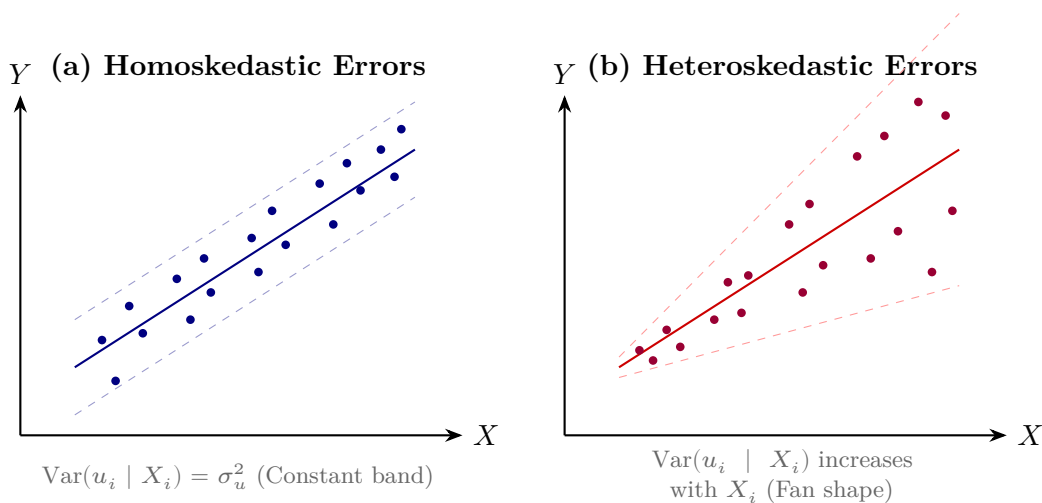


Figure 1: Homoskedastic vs. Heteroskedastic error distributions around the regression line.

Under homoskedasticity, we can simplify the numerator variance σ_v^2 :

$$\begin{aligned}
 \sigma_v^2 &= \mathbb{E}[(X_i - \mu_X)^2 u_i^2] \\
 &= \mathbb{E}[(X_i - \mu_X)^2 \mathbb{E}[u_i^2 | X_i]] \quad (\text{by LIE}) \\
 &= \mathbb{E}[(X_i - \mu_X)^2 \sigma_u^2] \quad (\text{since } \text{Var}(u_i | X_i) = \sigma_u^2) \\
 &= \sigma_u^2 \mathbb{E}[(X_i - \mu_X)^2] \\
 &= \sigma_u^2 \sigma_X^2
 \end{aligned}$$

Substituting $\sigma_v^2 = \sigma_u^2 \sigma_X^2$ into the general variance formula:

$$\text{Var}_{\text{homo}}(\hat{\beta}_1) = \frac{\sigma_u^2 \sigma_X^2}{n (\sigma_X^2)^2} = \frac{\sigma_u^2}{n \sigma_X^2}$$

$$\text{Var}_{\text{homo}}(\hat{\beta}_1) = \frac{\sigma_u^2}{n \sigma_X^2}$$

2. The Heteroskedastic Case (General Case)

Errors are **heteroskedastic** if the conditional variance depends on X_i :

$$\text{Var}(u_i | X_i) = \sigma_u^2(X_i) \neq \text{constant}$$

In this case, we cannot factor out σ_u^2 from the expectation, so $\sigma_v^2 \neq \sigma_u^2 \sigma_X^2$. We must retain the full

heteroskedasticity-robust variance formula:

$$\text{Var}_{\text{robust}}(\hat{\beta}_1) = \frac{1}{n} \frac{\sigma_v^2}{\sigma_X^4} = \frac{1}{n} \frac{\text{Var}((X_i - \mu_X)u_i)}{(\text{Var}(X_i))^2}$$

Estimating the Variances in Practice: Standard vs. Robust Estimators

The population formulas derived above depend on the **unobserved errors** u_i and the **unknown population moments** $\sigma_u^2, \sigma_v^2, \sigma_X^2$. In empirical applications, we estimate these variances by replacing u_i with the **OLS residuals** $\hat{u}_i = Y_i - \hat{Y}_i$ and population moments with their sample counterparts:

1. **Standard (Homoskedastic) Variance Estimator:** Assuming homoskedasticity, we estimate the error variance σ_u^2 using the squared **Standard Error of the Regression (SER)**:

$$s_u^2 \equiv \text{SER}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2 = \frac{\text{SSR}}{n-2}$$

Substituting s_u^2 into $\text{Var}_{\text{homo}}(\hat{\beta}_1) = \frac{\sigma_u^2}{\sum_{i=1}^n (X_i - \bar{X})^2}$ gives the **standard (classical) variance estimator**:

$$\widehat{\text{Var}}_{\text{homo}}(\hat{\beta}_1) = \frac{s_u^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \Rightarrow \text{SE}_{\text{homo}}(\hat{\beta}_1) = \frac{s_u}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2}}$$

Limitation: This estimator is valid **only if** the homoskedasticity assumption holds. If errors are heteroskedastic, $\widehat{\text{Var}}_{\text{homo}}(\hat{\beta}_1)$ is biased and inconsistent, causing test statistics to be invalid.

2. **Heteroskedasticity-Robust Variance Estimator (Eicker-Huber-White / Sandwich Estimator):** Without assuming homoskedasticity, we directly estimate $\sigma_v^2 = \text{Var}((X_i - \mu_X)u_i)$ using the residuals $\hat{v}_i = (X_i - \bar{X})\hat{u}_i$. This yields the **heteroskedasticity-robust variance estimator**:

$$\widehat{\text{Var}}_{\text{robust}}(\hat{\beta}_1) = \frac{\sum_{i=1}^n (X_i - \bar{X})^2 \hat{u}_i^2}{[\sum_{i=1}^n (X_i - \bar{X})^2]^2} \Rightarrow \text{SE}_{\text{robust}}(\hat{\beta}_1) = \frac{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \hat{u}_i^2}}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Advantage: This estimator is **consistent whether the errors are homoskedastic or heteroskedastic**. Because heteroskedasticity is common in economic data, modern empirical research almost universally defaults to robust standard errors.

Hypothesis Testing with the t -Statistic: With an estimator for the standard error $\text{SE}(\hat{\beta}_1)$ in hand (ideally heteroskedasticity-robust), we can now perform statistical inference as usual. Under the null hypothesis $H_0 : \beta_1 = \beta_{1,0}$ (most commonly $H_0 : \beta_1 = 0$),

the t -statistic:

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{\text{SE}(\hat{\beta}_1)} \stackrel{a}{\sim} \mathcal{N}(0, 1)$$

is asymptotically standard normal by the Central Limit Theorem. We can therefore carry out hypothesis tests (e.g., rejecting H_0 at the 5% significance level if $|t| > 1.96$), compute p -values, and construct asymptotic confidence intervals:

$$\hat{\beta}_1 \pm z_{1-\alpha/2} \times \text{SE}(\hat{\beta}_1)$$

in the standard fashion.

2 Omitted Variable Bias (OVB)

Least Squares Assumption 1 requires that the error term and the regressor are uncorrelated:

$$\text{Corr}(X_i, u_i) = 0 \quad (\text{or } \mathbb{E}[u_i | X_i] = 0)$$

When this assumption is violated, the OLS estimator is **biased and inconsistent**. The leading cause of this violation in observational data is **Omitted Variable Bias (OVB)**.

2.1 General Framework

Consider the linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

where u_i collects all other unobserved factors that influence Y_i .

If there is an omitted variable Z_i that affects Y_i , that variable is **hiding in the error term** u_i . Suppose we estimate the single-variable model of Y_i on X_i by OLS, ignoring Z_i .

Recall the formula for the OLS estimation error:

$$\hat{\beta}_1 - \beta_1 = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u})}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2}$$

In large samples (by the Law of Large Numbers):

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(u_i - \bar{u}) \xrightarrow{p} \text{Cov}(X_i, u_i) = \sigma_{Xu}$$

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \xrightarrow{p} \text{Var}(X_i) = \sigma_X^2$$

Therefore:

$$\hat{\beta}_1 - \beta_1 \xrightarrow{p} \frac{\sigma_{Xu}}{\sigma_X^2}$$

Using the definition of correlation $\rho_{Xu} = \text{Corr}(X_i, u_i) = \frac{\sigma_{Xu}}{\sigma_X \sigma_u} \implies \sigma_{Xu} = \sigma_X \sigma_u \rho_{Xu}$, we rewrite the ratio:

$$\frac{\sigma_{Xu}}{\sigma_X^2} = \frac{\sigma_X \sigma_u \rho_{Xu}}{\sigma_X^2} = \left(\frac{\sigma_u}{\sigma_X} \right) \rho_{Xu}$$

This gives the fundamental **Omitted Variable Bias** formula:

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \left(\frac{\sigma_u}{\sigma_X} \right) \rho_{Xu}$$

2.2 The Two Conditions for Omitted Variable Bias

Because standard deviations are strictly positive ($\sigma_u > 0$ and $\sigma_X > 0$), $\hat{\beta}_1$ is biased and inconsistent whenever $\rho_{Xu} \neq 0$.

An omitted factor Z creates omitted variable bias if and only if **both** conditions hold:

1. **Z is a determinant of Y :** Z affects Y , meaning it is part of the error term u .
 2. **Z is correlated with X :** $\text{Corr}(X, Z) \neq 0$, which makes X correlated with the error term u ($\rho_{Xu} \neq 0$).
- If Z is not a determinant of Y , it does not enter u .
 - If Z is uncorrelated with X ($\text{Corr}(X, Z) = 0$), then $\rho_{Xu} = 0$, and $\hat{\beta}_1 \xrightarrow{p} \beta_1$ remains unbiased and consistent.

When both conditions hold, $\rho_{Xu} \neq 0$, and the sign of the bias is determined by the sign of ρ_{Xu} :

$$\text{sgn}(\text{Bias}) = \text{sgn}(\rho_{Xu})$$

Determining the Direction of Bias

Suppose the true relationship is given by the two-variable model $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \tilde{u}_i$. If we omit X_{2i} and run the simple regression of Y_i on X_{1i} , the omitted variable X_{2i} enters the error term ($u_i = \beta_2 X_{2i} + \tilde{u}_i$).

The correlation between X_{1i} and the error term is then determined by the product of the effect of X_{2i} on Y_i (β_2) and the correlation between the two regressors:

$$\text{sgn}(\text{Bias}) = \text{sgn}(\beta_2) \times \text{sgn}(\text{Corr}(X_{1i}, X_{2i}))$$

	$\text{Corr}(X_{1i}, X_{2i}) > 0$	$\text{Corr}(X_{1i}, X_{2i}) < 0$
$\beta_2 > 0$ (X_2 positively affects Y)	Positive Bias ($\hat{\beta}_1$ overstates β_1)	Negative Bias ($\hat{\beta}_1$ understates β_1)
$\beta_2 < 0$ (X_2 negatively affects Y)	Negative Bias ($\hat{\beta}_1$ understates β_1)	Positive Bias ($\hat{\beta}_1$ overstates β_1)

The Solution to OVB: Multiple Regression

The direct econometric remedy for omitted variable bias is **Multiple Regression**: explicitly include the omitted confounding variable in the regression model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

By including X_{2i} , it is removed from the error term u_i . The coefficient β_1 now estimates the **partial effect** (or *ceteris paribus* effect) of X_{1i} on Y_i , **holding X_{2i} constant**.

3 Multiple Regression & Multicollinearity

While multiple regression solves omitted variable bias, introducing multiple regressors gives rise to a critical inference issue: **Multicollinearity**.

When evaluating the OLS estimator $\hat{\beta}_1$ for a variable of interest X_1 , we face two distinct econometric goals:

1. **Unbiasedness & Consistency** ($\mathbb{E}[\hat{\beta}_1] = \beta_1$, $\hat{\beta}_1 \xrightarrow{p} \beta_1$):
 - *Primary Threat*: **Omitted Variable Bias (OVB)**.
 - *Solution*: **Multiple Regression** (conditioning on observable confounders).
2. **Accurate Standard Errors & Statistical Precision**:
 - *Threat 1*: **Heteroskedasticity** $\implies$ *Solution*: Heteroskedasticity-robust standard errors.
 - *Threat 2*: **Multicollinearity** $\implies$ *Consequence*: Inflated standard errors ($\text{SE}(\hat{\beta}_1) \uparrow$), deflated t -statistics, and loss of statistical power (failure to reject $H_0 : \beta_1 = 0$).

3.1 Least Squares Assumptions for Multiple Regression

For the multiple regression model with k regressors:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i, \quad i = 1, \dots, n$$

the core **Least Squares Assumptions (LSAs)** are:

Least Squares Assumptions (LSA 1–4):

1. LSA 1 (Conditional Mean Zero / Exogeneity):

$$\mathbb{E}[u_i \mid X_{1i}, X_{2i}, \dots, X_{ki}] = 0 \implies \text{Corr}(X_{ji}, u_i) = 0 \quad \text{for all } j = 1, \dots, k$$

2. LSA 2 (i.i.d. Sampling):

$(X_{1i}, X_{2i}, \dots, X_{ki}, Y_i), i = 1, \dots, n$ are independently and identically distributed (i.i.d.)

3. LSA 3 (Large Outliers are Rare):

$$\mathbb{E}[X_{ji}^4] < \infty \quad \text{and} \quad \mathbb{E}[Y_i^4] < \infty \quad \text{for all } j = 1, \dots, k$$

4. LSA 4 (No Perfect Multicollinearity): No regressor is constant, and **no regressor is a perfect linear function of other regressors.**

Under LSAs 1–4, the OLS estimators are **unbiased** and **consistent**. Under homoskedasticity (LSA 5), the **Gauss-Markov Theorem** establishes that OLS is the **Best Linear Unbiased Estimator (BLUE)**.

3.2 Perfect Multicollinearity

Definition & Theoretical Breakdown

Perfect multicollinearity occurs when one of the regressors is an exact linear function of other regressors:

$$X_{ki} = c_0 + c_1 X_{1i} + c_2 X_{2i} + \cdots + c_{k-1} X_{k-1,i} \quad \text{for all } i$$

where $c_0, c_1, \dots, c_{k-1}$ are constants.

The slope coefficient β_k measures the partial effect of varying X_k **while holding all other regressors constant**. If X_k is an exact linear combination of the other regressors, it is logically impossible for X_k to vary while keeping the other regressors fixed. The OLS formulas involve division by zero (the normal equations have infinitely many solutions). The OLS estimators are **undefined and cannot be computed**.

A common cause of perfect multicollinearity is the **Dummy Variable Trap**. When a categorical variable with m mutually exclusive and exhaustive categories is represented by m indicator

variables:

$$D_{1i} + D_{2i} + \dots + D_{mi} = 1 \quad \text{for all } i$$

If an intercept β_0 and all m dummy variables are included simultaneously, the sum of the dummies perfectly matches the constant regressor (1).

Resolutions:

1. **Omit One Dummy Variable (The Benchmark / Reference Category):** Include $m-1$ dummies and an intercept β_0 . The intercept becomes the expected value of Y for the omitted category, and each dummy coefficient measures the differential relative to the base group.
2. **Omit the Intercept:** Include all m dummies and suppress the intercept ($\beta_0 = 0$). Each coefficient directly equals the group mean.

3.3 Imperfect Multicollinearity

Definition & Crucial Distinction

Imperfect multicollinearity occurs when two or more regressors are **highly, but not perfectly, correlated** (e.g., $|\text{Corr}(X_1, X_2)|$ is high).

- Imperfect multicollinearity is **NOT a violation of LSA 4**.
- **OLS remains unbiased, consistent, and BLUE:**

$$\mathbb{E}[\hat{\beta}_j] = \beta_j \quad \text{and} \quad \hat{\beta}_j \xrightarrow{p} \beta_j$$

- It is not an analytical flaw of the estimator; it is an inherent **data / sample precision problem** (*micronumerosity*—having limited independent information in the sample to separate two entangled variables).

The Multiple Regression Variance Formula

For the two-regressor model $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$ under homoskedasticity:

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma_u^2}{n \sigma_{X_1}^2 (1 - \rho_{12}^2)}$$

where $\rho_{12} = \text{Corr}(X_{1i}, X_{2i})$.

In the general k -regressor model:

$$\text{Var}(\hat{\beta}_j) = \frac{1}{1 - R_j^2} \times \frac{\sigma_u^2}{n \sigma_{X_j}^2} = \text{VIF}_j \times \frac{\sigma_u^2}{n \sigma_{X_j}^2}$$

where R_j^2 is the R^2 from regressing X_j on all other regressors, and $VIF_j \equiv \frac{1}{1-R_j^2}$ is the **Variance Inflation Factor**.

Intuition

To estimate the partial effect β_1 holding X_2 constant, OLS uses only the variation in X_1 that is **independent** of X_2 .

As the correlation between regressors approaches 1:

$$|\rho_{12}| \rightarrow 1 \implies (1 - \rho_{12}^2) \rightarrow 0 \implies \text{Var}(\hat{\beta}_1) \rightarrow \infty \implies \text{SE}(\hat{\beta}_1) \rightarrow \infty$$

When regressors move tightly together, the sample contains almost no observations where X_1 varies while X_2 stays constant. Without independent variation, OLS cannot separate their individual effects, causing the **sampling variance to explode**.

Consequences for Hypothesis Testing & Inference

1. **Inflated Standard Errors:** $\text{SE}(\hat{\beta}_1) = \sqrt{\text{Var}(\hat{\beta}_1)}$ becomes very large.
2. **Deflated t -Statistics:**

$$t = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)} \downarrow$$

Because the inflated standard error sits in the denominator, the t -statistic collapses toward zero.

3. **Loss of Statistical Power (Type II Error):** Even if X_1 has a true nonzero effect ($\beta_1 \neq 0$), the empirical t -statistic is unlikely to exceed the critical value ($|t| < 1.96$). We fail to reject $H_0 : \beta_1 = 0$, falsely concluding the variable is “insignificant.”
 4. **Wide Confidence Intervals:** The interval $\hat{\beta}_1 \pm 1.96 \times \text{SE}(\hat{\beta}_1)$ becomes too wide to provide useful guidance.
 5. **Classic Diagnostic Symptom:** The overall regression has a high R^2 and a statistically significant joint F -test, yet the individual regressors have statistically insignificant t -statistics.
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The OVB vs. Multicollinearity Trade-off

When faced with inflated standard errors and low t -statistics, researchers must **never drop a relevant confounding variable** just to make remaining variables statistically significant:

- **Multicollinearity affects precision only:** The OLS estimator remains **unbiased and consistent**.

- **Dropping a relevant variable creates Omitted Variable Bias:** Violates LSA 1 ($\mathbb{E}[u | X] \neq 0$), making the estimator **biased and inconsistent**.
- **Econometric Principle: Never trade bias for variance.** An unbiased estimator with a wide confidence interval is far superior to a biased estimator that is precisely wrong.

Legitimate Remedies:

1. **Increase Sample Size (n):** Because $\text{Var}(\hat{\beta}_1) \propto \frac{1}{n}$, collecting a larger sample directly shrinks standard errors and overcomes multicollinearity.
 2. **Joint Hypothesis Testing (F -Tests):** Test the joint significance of collinear variables ($H_0 : \beta_1 = 0$ and $\beta_2 = 0$). The joint F -test often detects statistical significance even when individual t -tests cannot.
 3. **Collinear Control Variables:** Collinearity between two control variables included solely to prevent OVB does not inflate the standard error of the policy variable of interest, as long as the policy variable itself is not collinear with them.
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4 Measures of Fit in Multiple Regression

4.1 The Standard Error of the Regression (SER)

The SER estimates the standard deviation of the error term u_i . In multiple regression, we adjust the degrees of freedom for the $k + 1$ estimated parameters ($\beta_0, \beta_1, \dots, \beta_k$):

$$s_u \equiv \text{SER} = \sqrt{\frac{\text{SSR}}{n - k - 1}} = \sqrt{\frac{\sum_{i=1}^n \hat{u}_i^2}{n - k - 1}}$$

where $n - k - 1$ is the residual degrees of freedom. The SER is measured in the exact same units as the dependent variable Y .

4.2 R^2 in Multiple Regression

Recall the definition of the coefficient of determination:

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{SSR}}{\text{TSS}} = 1 - \frac{\sum_{i=1}^n \hat{u}_i^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2}$$

- **Mathematical Property:** R^2 **never decreases** (and almost always increases) whenever an additional regressor is added to the model, **even if that regressor is completely unrelated to Y or pure noise**.
- **Why?** OLS chooses coefficients to minimize the Sum of Squared Residuals (SSR). Adding another regressor expands the set of choices over which OLS minimizes SSR. At worst, OLS

can assign a coefficient of zero ($\hat{\beta}_{k+1} = 0$), so $\text{SSR}_{\text{new}} \leq \text{SSR}_{\text{old}}$. Because TSS depends only on Y and remains fixed, $1 - \frac{\text{SSR}}{\text{TSS}}$ can never fall.

- **Consequence:** R^2 cannot be used to evaluate whether adding a variable improves model specification.

4.3 The Adjusted R^2 ($\bar{R}^2$)

To fix this defect, the **Adjusted R^2** (denoted $\bar{R}^2$) penalizes the inclusion of additional regressors by dividing each sum of squares by its corresponding degrees of freedom:

$$\bar{R}^2 = 1 - \frac{\text{SSR}/(n-k-1)}{\text{TSS}/(n-1)} = 1 - \left(\frac{n-1}{n-k-1} \right) \frac{\text{SSR}}{\text{TSS}}$$

where:

- n : sample size
- k : number of regressors (excluding the constant term)
- $\frac{n-1}{n-k-1} > 1$: the **degrees-of-freedom penalty factor** (which increases as more regressors k are added)

Alternatively, because $s_u^2 = \frac{\text{SSR}}{n-k-1}$ is the sample variance of the residuals and $s_Y^2 = \frac{\text{TSS}}{n-1}$ is the sample variance of Y :

$$\bar{R}^2 = 1 - \frac{s_u^2}{s_Y^2}$$

$\bar{R}^2$ can be negative if the regressors explain very little variation in Y and the penalty factor pushes the subtracted term above 1.